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Read the following passage.

Lake Malawi in East Africa has more species of fish than any other lake in the world. Many of these species have evolved from a common ancestor. Lake Malawi is one of the largest lakes in the world and was formed several million years ago. Since then, the water level has fluctuated greatly. As a result, what is now a large lake was at one time many smaller, separate lakes.

5

The country of Malawi has a total area of 118 000 km². The actual land area is only 94 080 km², because approximately one-fifth of the country is Lake Malawi.

In December 1990, forests covered 41.4% of the actual land area of Malawi.
In December 2016, forests covered 26.4% of the actual land area of Malawi.

10

Deforestation and farming along the shores of Lake Malawi have caused increased soil erosion and loss of nutrients into the lake. This has resulted in a decrease in some fish populations. The mark-release-recapture method can be used to estimate the size of a fish population. However, this method can produce unreliable results in very large lakes.

15

Use the information in the passage and your own knowledge to answer the following questions.

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Lake Malawi in East Africa has more species of fish than any other lake in the world (line 1).

Suggest and explain how this speciation may have occurred.

[4 marks]

[illegible]

Calculate the mean loss of forest cover in km² per week during this time period.

Answer km² per week

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Explain why.

[4 marks]

[illegible]

The mark-release-recapture method can be used to estimate the size of a fish population (lines 13–14).

[4 marks]

[illegible]

Suggest why the mark-release-recapture method can produce unreliable results in very large lakes (lines 14–15).

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